

Lecture of the week:

- *Using's Meiss's action principle to quantify chaos in fusion devices.*

Rais, L. (2025). Master Thesis, EPFL

- *Turnstile flux as a measure for chaotic transport in magnetic confinement fusion devices.*

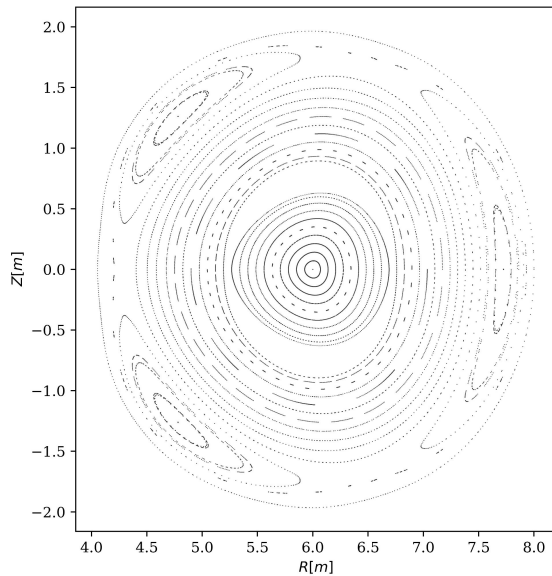
Smiet et al. (2025). Chaos: An Interdisciplinary Journal of Nonlinear Science.

- *Thirty years of Turnstiles and Transport.*

Meiss, J.D. (2015). Department of Applied Mathematics, University of Colorado

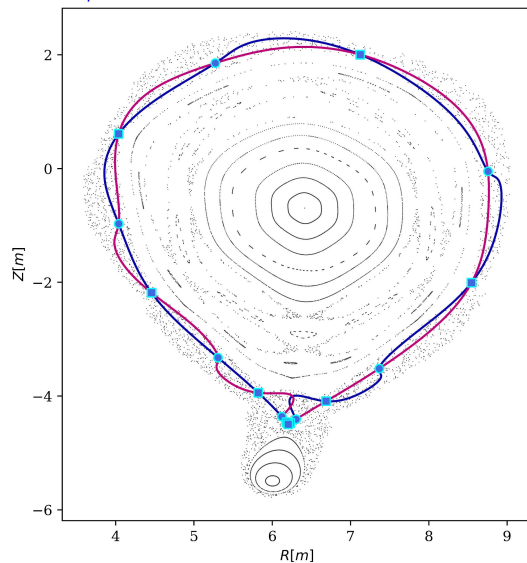
First numerical steps:

- Setup GitHub, Vs code and Pyoculus package
- Familiarization with Pyoculus code
- First Poincaré plots with different types of perturbation



Maxwell-Boltzmann pert. ($m/n=3/2$)

Turnstile flux: $\phi = 1.601e+00$



MB + circ. current loop perturbation
with homoclinic points